

# **REPORT**

## **Title**

### **Implementation of the Vacuum Cleaner Problem Using Python**

## **Aim**

To implement a Vacuum Cleaner Agent in Python that cleans all dirty rooms in the environment.

## **Problem Statement**

The Vacuum Cleaner Problem is a simple Artificial Intelligence problem in which an intelligent agent operates in an environment containing multiple rooms. Each room can be either clean or dirty. The agent senses the condition of its current room and performs appropriate actions such as cleaning the room or moving to another room. The objective is to ensure that all rooms become clean while minimizing unnecessary actions. The program should simulate the behavior of the vacuum cleaner agent and display the actions taken to clean the environment.

## **Requirements**

- Python 3.x

## **Procedure**

1. Represent the rooms using a list.

2. Assign each room a status of **Clean** or **Dirty**.
3. Place the vacuum cleaner at the first room.
4. Check the status of the current room.
5. If the room is dirty, clean it; otherwise, move to the next room.
6. Repeat the process until all rooms have been visited.
7. Display the action performed for each room.
8. Display the final status of all rooms after cleaning.

## Program

```
def vacuum_cleaner(rooms):  
    for i in range(len(rooms)):  
        print(f"Room {i+1}: {rooms[i]}")  
  
        if rooms[i] == "Dirty":  
            print("Action: Clean")  
            rooms[i] = "Clean"  
        else:  
            print("Action: No Cleaning Required")  
  
    print()  
  
    print("Final Room Status:")  
    for i in range(len(rooms)):  
        print(f"Room {i+1}: {rooms[i]}")  
  
rooms = ["Dirty", "Clean", "Dirty", "Dirty"]  
  
vacuum_cleaner(rooms)
```

## Output

```
Room 1: Dirty  
Action: Clean
```

```
Room 2: Clean  
Action: No Cleaning Required
```

Room 3: Dirty  
Action: Clean

Room 4: Dirty  
Action: Clean

Final Room Status:  
Room 1: Clean  
Room 2: Clean  
Room 3: Clean  
Room 4: Clean

## **Result**

The Vacuum Cleaner Problem was successfully implemented in Python. The agent correctly identified dirty rooms, cleaned them, and produced a final environment in which all rooms were clean.

## **Conclusion**

The Vacuum Cleaner Problem illustrates the behavior of a **simple reflex agent** in Artificial Intelligence. The agent perceives the environment, performs actions based on the current state, and achieves the goal of cleaning all rooms. This problem demonstrates fundamental AI concepts such as intelligent agents, perception, action, state representation, and problem-solving in a simplified environment.